



Quantification of rare earth elements using laser-induced breakdown spectroscopy☆

Madhavi Martin^{a,*}, Rodger C. Martin^b, Steve Allman^a, Deanne Brice^c, Ann Wymore^a, Nicolas Andre^d

^a Biological Sciences Division, Oak Ridge, TN 37831, USA

^b Fusion and Materials for Nuclear Systems Division, Oak Ridge, TN 37831, USA

^c Environmental Sciences Division, Oak Ridge National Laboratory, Oak Ridge, TN 37831, USA

^d The University of Tennessee, Center for Renewable Carbon, Knoxville, TN 37996, USA

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ABSTRACT

A study of the optical emission as a function of concentration of laser-ablated yttrium (Y) and of six rare earth elements, europium (Eu), gadolinium (Gd), lanthanum (La), praseodymium (Pr), neodymium (Nd), and samarium (Sm), has been evaluated using the laser-induced breakdown spectroscopy (LIBS) technique. Statistical methodology using multivariate analysis has been used to obtain the sampling errors, coefficient of regression, calibration, and cross-validation of measurements as they relate to the LIBS analysis in graphite-matrix pellets that were doped with elements at several concentrations. Each element (in oxide form) was mixed in the graphite matrix in percentages ranging from 1% to 50% by weight and the LIBS spectra obtained for each composition as well as for pure oxide samples. Finally, a single pellet was mixed with all the elements in equal oxide masses to determine if we can identify the elemental peaks in a mixed pellet. This dataset is relevant for future application to studies of fission product content and distribution in irradiated nuclear fuels. These results demonstrate that LIBS technique is inherently well suited for the future challenge of in situ analysis of nuclear materials. These studies also show that LIBS spectral analysis using statistical methodology can provide quantitative results and suggest an approach in future to the far more challenging multielemental analysis of ~20 primary elements in high-burnup nuclear reactor fuel.

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1. Introduction

In nuclear fuel fission reactions, each uranium or plutonium atom will split into two lighter elements. Rare earth elements are a major component of these fission products. Since fission product content in nuclear fuels is proportional to the fraction of “burned” fuel, it can be used to estimate actinide masses (U, Pu, etc.) for nonproliferation “safeguards” monitoring. The reprocessing of irradiated fuel must remove most fission products prior to reuse as fuel. Therefore, it is important to monitor the presence of mixed actinide, rare earths, and other fission product elements throughout the separation process. Challenges in monitoring fission product content using conventional analytical techniques can be significant, such as in electrochemical recycling processes [1]. The emission spectra of rare earths are very complex and spectral interferences must be known, so a detailed and comprehensive study of the predominant rare earths using LIBS has been undertaken in this research and reported here. There are a number of detailed research articles that have shown the LIBS

measurements on actinides, their surrogates, [2–7,27] and rare earth elements [8–26]. This study is the most comprehensive LIBS analysis of rare earth elements individually and in a mixture.

Prior research shows that the technique of LIBS has been used in the elemental analysis of the rare earth elements [8–27]. Among those, europium has been studied the most. Eu⁺ has been detected in colloidal solutions and the technique has been used in the determination of solubility data and of concentrations of europium in glass matrices [8–13]. Another study has determined trace levels of samarium, europium, and gadolinium in aqueous samples by LIBS [9]. Stainless steel-coated gadolinium and gadolinium in its solid oxide form and in molten glass matrices have been studied [15,16]. LIBS studies of other rare earths such as lanthanum [14,17], neodymium [18–22], praseodymium, samarium [27,23], and yttrium [24–27] have been also been performed.

The rare earths studied here using the LIBS technique are europium (Eu), gadolinium (Gd), lanthanum (La), neodymium (Nd), praseodymium (Pr), samarium (Sm), and the transition metal yttrium (Y). The rationale for selecting these particular elements for the LIBS studies in this article is given by a typical example: in irradiated nuclear reactor fuel, after decay of 150 days, ~18% of the fissions result in Nd atoms, 13% in Ce atoms (previously studied using LIBS [28]), 6% in La and in Pr, 4% in Y and in Sm, 0.6% in Eu, and 0.3% in Gd [29]. There are numerous difficulties associated with undertaking the simultaneous analysis of

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* Corresponding author at: Biological Sciences Division, Oak Ridge National Laboratory, Oak Ridge, TN 37831, USA.

actinides, rare earths, and other fission products, and extremely dense and complex spectral features result from the as-produced mixture. These difficulties are (i) experimental, (ii) operational, and (iii) safety related. The technique of choice for measuring the actinide and rare earth content in irradiated fuel is ICP-MS or ICP-OES/AES [30–41], but the challenge of analyzing concentrated solutions from fuel dissolution with very high radiological dose must first be addressed. This creates analytical uncertainties from large dilutions, up to a million-fold, which can be costly and time consuming and increase worker hazards in sample handling. Furthermore, complex spectral features can create problems in identification, fingerprinting, and in the quantification of these elements. Operational difficulties often involve burdensome paperwork, approvals, and controls that are required when handling even < 1 g of fissile material such as enriched uranium or plutonium. Setting up glove boxes for radioanalytical work has significant operational costs, and hot cell costs are even higher. Pre-operational safety analysis of new equipment and operations and of radioactive content and its potential for release is far more rigorous than typical laser and chemical safety requirements. In addition to the experimental and operational difficulties, each nuclear facility in the U. S. has “safety basis documents” that set nuclear safety boundaries for experimental activities. Any new hazard introduced into a nuclear facility must be analyzed and documented. This work must be approved prior to beginning new experiments and can take months if not routine. For these reasons, LIBS has certain advantages as a micro-analytical technique in potentially mitigating some of these controls compared to conventional analytical techniques.

As with all analytical techniques, LIBS has inherent limitations that hinder quantification. Due to the matrix effects, true quantitative analyses and calibrations can be a problem. One drawback for the LIBS technique is that different matrices for different samples require matrix-specific calibration standards. Windom and Hahn have overcome the matrix effects for the LIBS technique in their article [42] and also discussions of the challenges for the quantitative analysis for the LIBS technique have been shown in the review articles by Hahn and Omenetto and other authors [43–45]. Certain issues such as sampling geometry on elemental emissions and other sampling parameters have been clearly addressed [46,47].

For the previously mentioned example of irradiated nuclear reactor fuel after decay of 150 days, the estimates of fission product content [29] indicate that ~23% of the fission product mass is represented by the seven elements measured in this work, which also provides a spectral database of these elements for further LIBS studies. A previous LIBS study of Cs, Sr, and Ce in the same matrix as here [28] covers another ~18% of this fission product mass and associated LIBS spectra. Separate LIBS studies of Mo, Zr (in U–Zr alloy form), and Pd contribute spectral data for another ~24% of the fission product mass. Spectral data from these separate studies can now be used to plan the LIBS measurement of and interpretation of results for nearly two-thirds of the fission product content (depending on the time of decay prior to measurement) in irradiated fuel samples. Synthesis of these spectral libraries with the spectral tabulations from recent mixed actinide studies [6] of U, Np, Pu, and Am now provides a starting point for analysis of complex nuclear

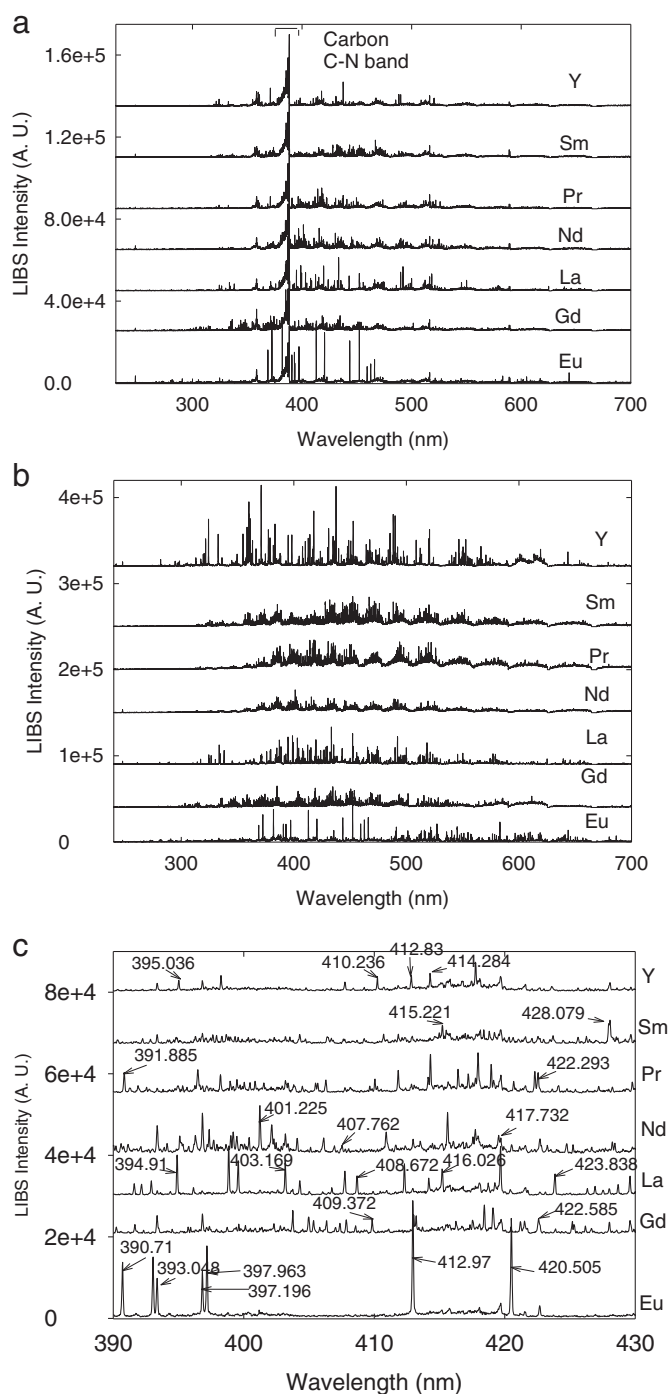


Fig. 1. (a) LIBS spectra for 1% concentration of Eu, Gd, La, Nd, Pr, Sm, and Y oxides individually in a graphite matrix. (b) LIBS spectra for 25% concentration of Eu, Gd, La, Nd, Pr, Sm, and Y oxides individually in a graphite matrix. (c) LIBS spectra for 25% concentrations of Eu–Gd, La, Nd, Pr, Sm, and Y oxides in the expanded wavelength window of 390–430 nm.

Table 1
Characteristics of laser and detector/spectrometer.

Laser characteristics		
Wavelength	Energy/pulse	Rep- rate
532 nm	45 mJ	10 Hz
Detector/spectrometer characteristics		Pixel resolution (FWHM) nm
Wavelength range nm	Practical spectral bandwidth	200–400 400–600 600–800
190–800	210–790 nm	0.06 0.06–0.08 0.09–0.12

fuel mixtures, including selection of optimal low-interference identifying peaks for each element.

In 2003, a feasibility experiment demonstrated LIBS analysis of the dried residue from 25 μ L of concentrated, undiluted radiochemical process solution containing significant masses of americium, curium, and rare earth and other fission products [48]. No ICP-based instrument could measure this sample without multiple orders-of-magnitude dilution and resultant propagation of analytical uncertainties. Since then, no comparable experiment has been reported in the literature, but the results clearly indicated the need for an *a priori* comprehensive

LIBS spectral library to qualitatively predict spectral interferences and optimize the selection of microvolume and mass content of the pipetted sample to facilitate subsequent data interpretation. Because of the prolific atomic emission lines from the major rare earth fission products (in addition to the even more prolific lines from any actinides present), this work represents a significant step toward experimental planning of future LIBS analyses of real-world nuclear samples.

The objectives of this article are as follows:

- To obtain the broadband LIBS emission spectra from single-element samples of six rare earth elements Eu, Gd, La, Nd, Pr, and Sm, and one transition metal Y, at concentrations of 1% and 50% by weight for each element in oxide form within a carbon matrix. This has allowed the authors to identify the specific peaks for each element and develop a calibration curve for each.
- To develop multivariate calibration and cross-validations for each individual element (shown in Table 3).
- To obtain the broadband LIBS emission spectrum from an equal mixture of the rare earth elements, Eu, Gd, La, Nd, Pr, and Sm, and one transition metal Y to see if individual emission lines from these can be identified in this mixture.

2. Experimental

LIBS is a laser-based technique in which, for this study, a laser of 532 nm wavelength is focused onto the surface of a sample which vaporizes and ionizes a small volume of sample material to

generate a plasma spark. All states of matter can be analyzed, e.g., solid, liquid, gaseous, and aerosol. LIBS vaporizes a small sample volume with sufficient energy to ionize all elemental components into optically excited states in the resultant sample plume. The vaporized species then undergo de-excitation and optical emission on a microsecond time scale, and time-dependent ultraviolet-visible spectroscopy fingerprints the elements associated with the spectral peaks. The plasma emits light that is collected by an optical fiber and delivered to a spectrometer. The spectrometer separates out the white light of the plasma into different wavelengths, and coupling onto the face of an intensified charge-coupled device (ICCD) detector converts the optical signal into an electronic signal. Powerful computer software is used to present a spectrum (intensity versus wavelength) of the sample under test. The process of excitation, light collection, separation into wavelengths, and optical signal conversion to electronic signal is completed within 100 milliseconds. LIBS is typically a surface analytical technique, with each laser pulse vaporizing microgram or submicrogram sample masses. However, the rapidity of sampling (typically 10 Hz laser repetition rate) and ability to scan a sample surface ablate a hole into a solid sample with repeated laser pulses (for depth profiling) or focus the laser spark below the surface of a liquid sample permits more versatile analyses and provides sufficient statistics for bulk sampling. The detection limits are often in the parts per million (1–100 ppm) range for most elements in the periodic table (dependent on sample matrix). These limits of detection can be reduced when the sample is placed under reduced atmospheric pressure or in an inert gaseous environment.

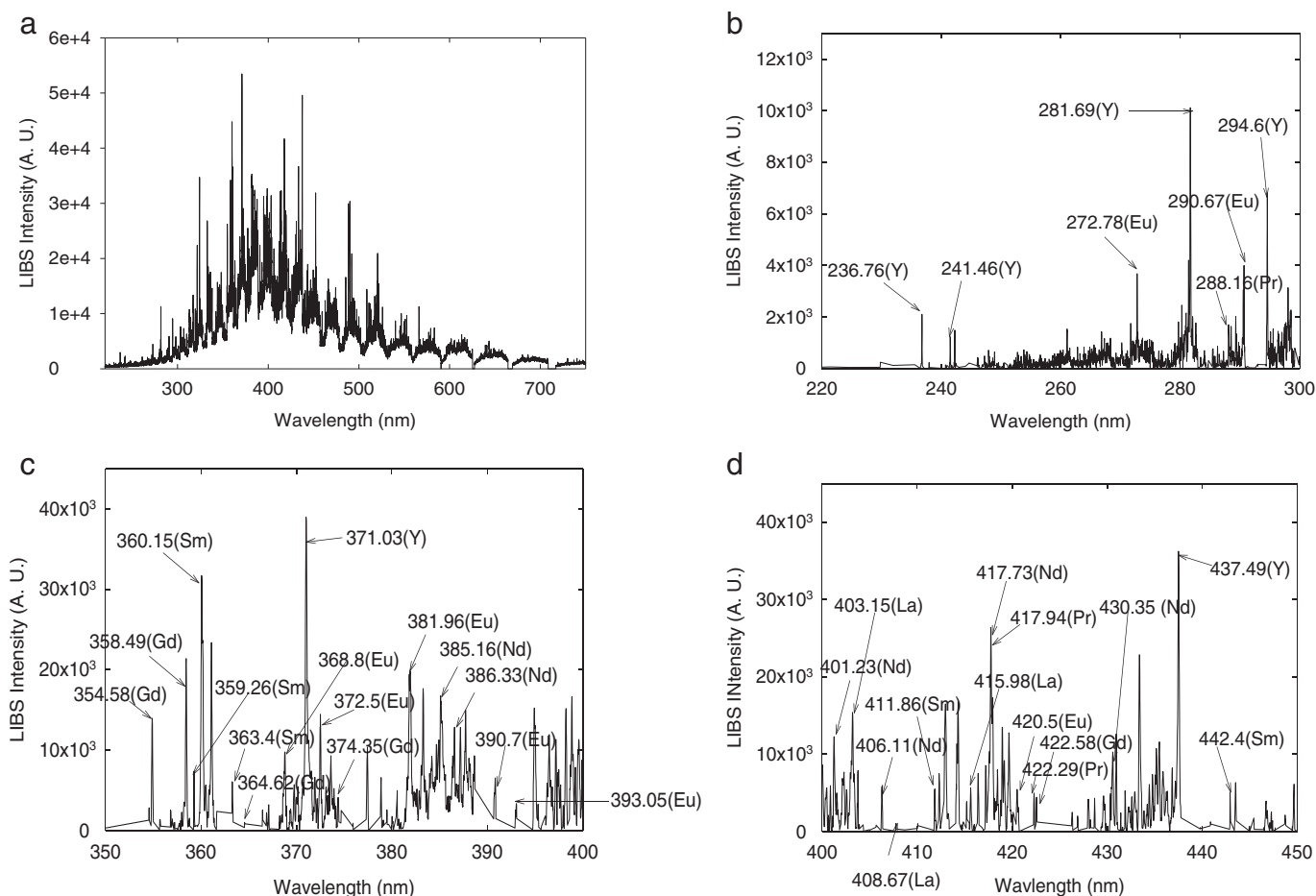


Fig. 2. (a) LIBS spectrum for a mixture of equal masses of Eu, Gd, La, Nd, Pr, Sm, and Y oxides. (b) LIBS spectrum for a mixture of equal masses of Eu, Gd, La, Nd, Pr, Sm, and Y oxides expanded into wavelength window from 220 to 300 nm. (c) LIBS spectrum for a mixture of equal masses of Eu, Gd, La, Nd, Pr, Sm, and Y oxides expanded into a wavelength window from 350 to 400 nm. (d) LIBS spectrum for a mixture of equal masses of Eu, Gd, La, Nd, Pr, Sm, and Y oxides expanded into a wavelength window from 400 to 450 nm.

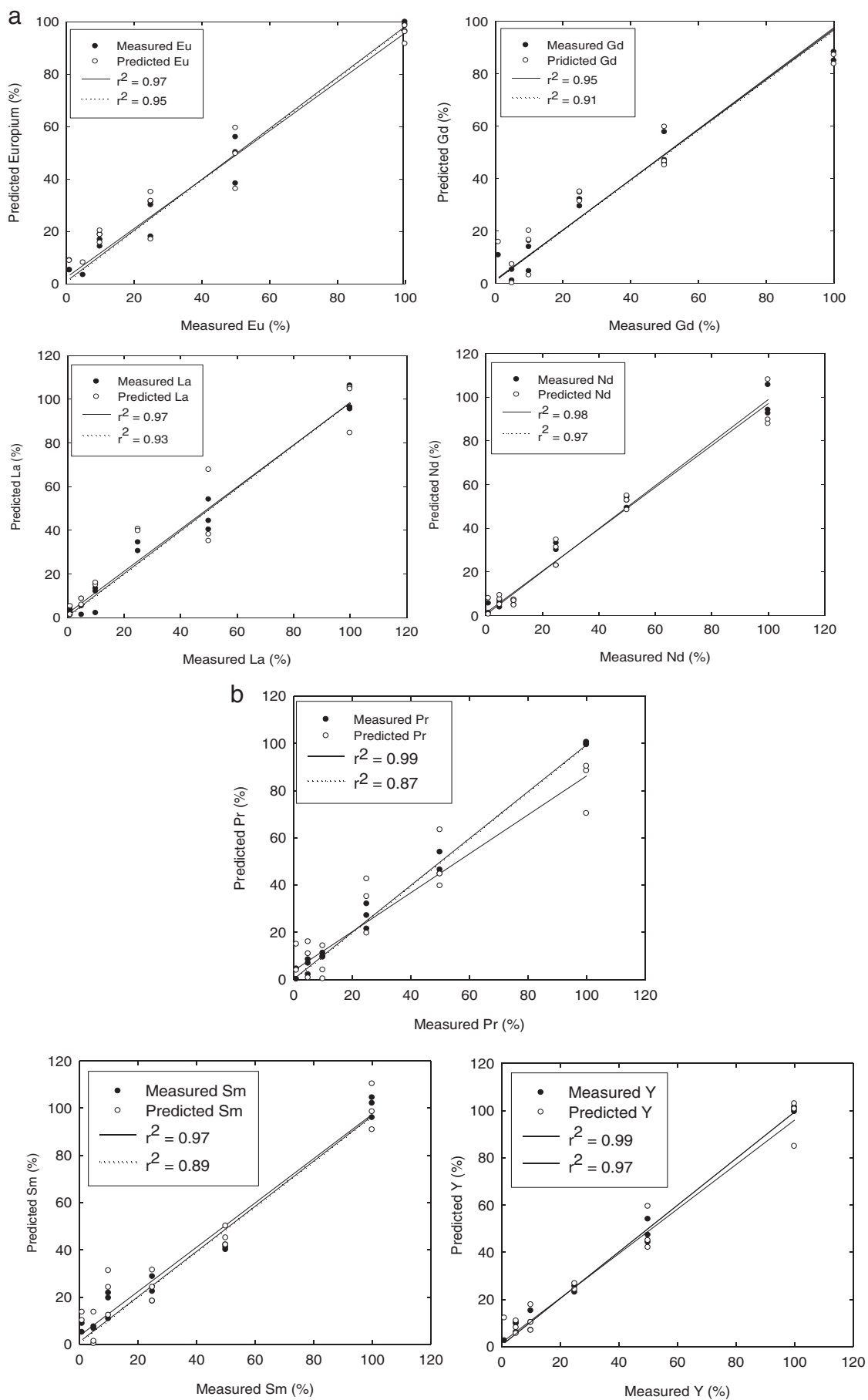
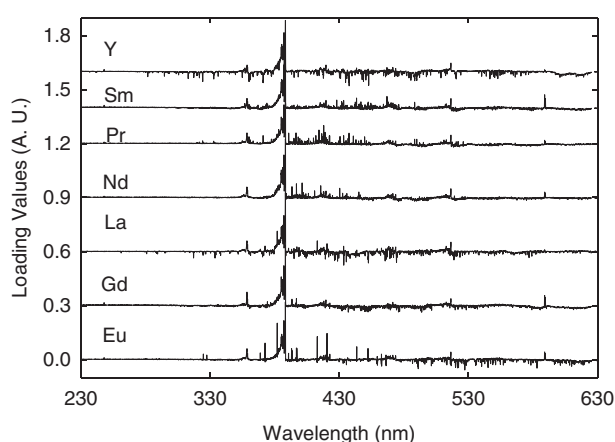


Table 2

Regression coefficients, offsets, and root mean square error (RMSE) for both calibration and validation sets for Eu, Gd, La, Nd, Pr, Sm, and Y.

Element	Slope	Offset	RMSE	R ²
Eu (calibration)	0.973	0.87	5.73	0.973
Eu (validation)	0.933	2.37	8.09	0.951
Gd (calibration)	0.954	1.47	7.43	0.954
Gd (validation)	0.967	1.06	10.4	0.913
La (calibration)	0.988	0.68	5.17	0.979
La (validation)	0.964	2.05	10.06	0.929
Nd (calibration)	0.985	0.48	4.27	0.985
Nd (validation)	0.958	1.27	6.04	0.973
Pr (calibration)	0.992	0.26	3.14	0.992
Pr (validation)	0.824	3.88	12.16	0.875
Sm (calibration)	0.969	1.02	6.30	0.968
Sm (validation)	0.920	1.97	12.25	0.896
Y (calibration)	0.993	0.22	2.88	0.993
Y (validation)	0.944	1.62	6.75	0.969

**Fig. 4.** The loading parameters for Eu, Gd, La, Nd, Pr, Sm, and Y.

2.1. Instrumentation

The main components of the experimental system are a pulsed Big Sky laser model CFR-Ultra which is a Q-switched Nd:YAG laser with frequency doubled output wavelength of 532 nm [28]. Laser energy of 45 mJ was used to obtain all the sample spectra. The experimental setup also has an X–Y–Z stage to provide capability of three-dimensional movements. A displacement laser (650 nm) is employed to control the vertical position of the sample upper surface, making sure that the laser beam which is exciting the sample surface remains focused and that the plasma is optimally located with respect to the collection optics. The light emitted by the plasma at the focal volume was collected by a set of collection optics and focused into a low O–H silica fiber. The optical fiber is used to deliver the light to an Echelle spectrometer from Catalina Scientific model SE 200 spectrometer that resolves light into different wavelengths by a high-order dispersion module (190–800 nm range), followed by detection using a 2-D 1024 × 1024 pixel intensified charge-coupled detector (ICCD) made by Andor Technologies. The advantage of using an ICCD is that the detection can be delayed with respect to plasma formation. The data were collected by using a delay of 1 microsecond, a gate width of 10 microseconds, and the repetition rate at which the laser was operated was 10 Hz. The whole process of sample excitation, plasma formation, light

Table 3

Loading parameters for the different elements identified by the calibration–validation methodology in the wavelength range 390–430 nm.

Element	Observed wavelength (nm)	Element	Observed wavelength (nm)	Element	Observed wavelength (nm)
Eu(II)	390.71	La(I)	416.026	Pr(II)	422.293
Eu(II)	392.887	La(II)	419.655	Sm(II)	392.828
Eu(II)	393.048	La(II)	423.838	Sm(II)	411.019
Eu(II)	397.196	Nd(II)	392.710	Sm(II)	411.855
Eu(II)	397.963	Nd(II)	395.745	Sm(II)	415.221
Eu(II)	412.97	Nd(II)	397.330	Sm(II)	419.945
Eu(I)	413.707	Nd(II)	397.949	Sm(II)	428.079
Eu(II)	420.505	Nd(II)	399.010	Y(II)	393.066
Gd(II)	396.929	Nd(II)	401.225	Y(II)	395.036
Gd(II)	404.684	Nd(II)	402.478	Y(II)	398.260
Gd(I)	405.364	Nd(II)	404.080	Y(I)	407.735
Gd(I)	405.822	Nd(II)	407.762	Y(I)	410.236
Gd(I)	407.870	Nd(II)	410.946	Y(I)	412.830
Gd(I)	409.372	Nd(II)	415.626	Y(I)	414.284
Gd(II)	418.425	Nd(II)	417.732	Y(II)	417.754
Gd(I)	419.078	Pr(II)	391.885	Y(II)	419.928
Gd(I)	422.585	Pr(II)	396.426		
La(II)	392.922	Pr(III)	398.051		
La(II)	394.91	Pr(II)	398.968		
La(II)	398.852	Pr(II)	399.585		
La(II)	399.575	Pr(II)	405.880		
La(II)	403.169	Pr(II)	410.072		
La(II)	404.291	Pr(II)	411.389		
La(II)	407.735	Pr(II)	414.311		
La(II)	408.672	Pr(II)	416.804		
La(II)	412.323	Pr(II)	417.225		
La(II)	415.197	Pr(II)	417.939		

collection, and optical-to-digital conversion of the signal takes at most a second to a few seconds depending on the number of shots that are averaged to get a good signal-to-noise ratio. Table 1 summarizes the laser and detector/spectrometer characteristics.

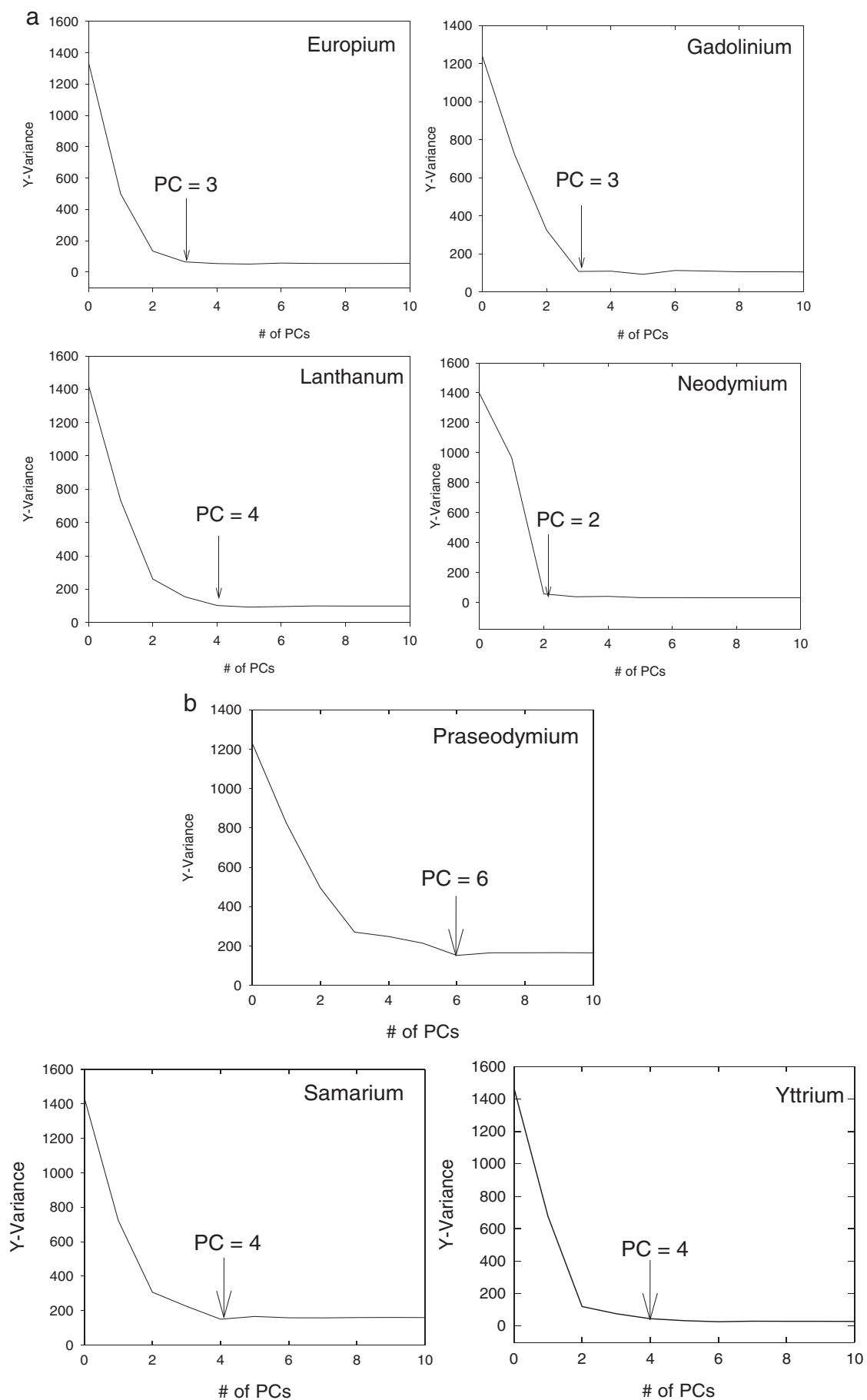
2.2. Reagents

All the rare earth oxide powders (Eu₂O₃, Gd₂O₃, La₂O₃, Nd₂O₃, Pr₆O₁₁, Sm₂O₃, and the transition metal oxide Y₂O₃), were obtained from Alpha Aesar, Ward Hill, Massachusetts (USA). The purity of the oxide powders that were used in this research was 99.99%. The graphite powder that was a natural, microcrystal grade, of 99.9995% purity was also obtained from Alpha Aesar, Ward Hill, Massachusetts (USA). The polyvinyl acetate (PVA) was 99–100% hydrolyzed and was obtained from Acros Organics, New Jersey, USA.

2.3. Preparation of samples

The powders of the rare earth oxides Eu₂O₃, Gd₂O₃, La₂O₃, Nd₂O₃, Pr₆O₁₁, Sm₂O₃, and the transition metal oxide Y₂O₃ were used in making the pellets. These powders were weighed along with a balancing amount of graphite powder diluent. The total weight of the oxide material and graphite mixture was 75 mg. These components were combined in a bottle and the mixture vortexed and stirred. A quantity of 300 µL of 0.5% PVA solution was pipetted into a glass tube and re-mixed a second time as in the previous step. A heating block was used to dry the mixed powders containing PVA. The contents of the bottle were emptied into a 1/4" die and pressed at 1500 lb for at least 1 minute. The pellets were placed in a labeled plastic bag until subsequent LIBS measurements.

Fig. 3. (a) r^2 —coefficient of determination for calibration and full cross-validation sample sets for Eu, Gd, La, and Nd. (b) r^2 —coefficient of determination for calibration and full cross-validation sample sets for Pr, Sm, and Y.



3. Results and discussions

LIBS spectra obtained using an Echelle spectrometer for 1% by weight of each of the rare earth (Eu, Gd, La, Nd, Pr, and Sm) and Y (transition metal) oxides, mixed with a 99% balance of graphite powder are shown in Fig. 1(a). The elemental concentrations range from 0.79% for Y to 0.87% for Gd. The acquired range of the spectra is 200–800 nm (reduced to 250–700 nm here), obtained under an argon purge atmosphere. The spectra are very complex with a dense distribution of spectral features that are extremely difficult and tedious to identify individually. Since the buildup of fission product elements from nuclear fission reactions can be as high as 18% of the total fissions for Nd, Fig. 1(b) shows the spectra of 25% by weight of each element's oxide (with elemental concentrations ranging from 19.7% for Y to 21.7% for Gd), in which the complexity and density of peaks increase rapidly over the 1% spectra shown in Fig. 1(a). By zooming into the spectral region of interest, the individual spectral lines can be clearly observed as shown in Fig. 1(c). Due to the density of the number of spectral lines, some line interferences are observed.

The main neutral and ionic lines of the rare earths have been identified in the expanded wavelength region (390–430 nm) in Fig. 1(c). This spectrum shows that multiple signature peaks for fingerprinting each rare earth element can be used in calibrating these individual elements. When we correlate the peaks in the LIBS spectra to the NIST-Atomic spectra data baselines, the peaks that were identified for Y are 393.066, 395.036, 398.26, 407.735, 410.236, 412.83, 414.284, 417.754, and 419.928 nm. The peaks identified for Sm are 392.828, 411.019, 411.855, 415.221, 419.945, and 428.079 nm. The peaks that were identified for Pr are 391.885, 396.426, 398.051, 398.968, 399.583, 405.88, 410.073, 411.389, 414.311, 416.804, 417.225, 417.939, and 422.293 nm. For Nd, the peaks of interests were 392.71, 395.745, 397.33, 397.949, 399.01, 401.225, 402.478, 404.08, 406.109, 407.762, 410.946, 415.626, and 417.732 nm. A number of peaks identified for La are 392.922, 394.91, 398.852, 399.575, 403.169, 404.291, 407.735, 408.672, 412.323, 415.197, 416.026, 419.655, and 423.838 nm. The peaks that were identified for Gd are 396.929, 404.684, 405.364, 405.822, 407.87, 409.372, 418.425, 419.078, and 422.585 nm; and lastly, the peaks correlated to Eu are 390.71, 392.887, 393.048, 397.196, 397.963, 412.97, 420.505 nm (Fig. 1(c)). All the peaks that are mentioned here have not been shown in Fig. 1(c) because of the scaling for the seven spectra it was difficult to label them individually. This shows that even in a small wavelength range, we have a cluster of 70 peaks that can be used to identify definitively the set of seven elements that have been studied for this article.

Fig. 2(a) shows the full broadband spectrum of an equal mixture of Eu, Gd, La, Nd, Pr, Sm, and Y oxides. The concentration of each oxide is 14.3% (elemental concentrations range from 11.3% Y to 12.4% Gd). After acquiring the data for this mixture, it was noticed that the complexity and richness of the spectral features increased greatly, as expected. To see if we can differentiate all seven rare earths within this complex mixture, the spectrum is expanded within the wavelength region of 220–300 nm in Fig. 2(b). After expanding the broadband spectrum to focus on a smaller window, it was observed that individual peaks of Eu, Pr, and Y could be resolved. Further investigation of the wavelength region between 350 and 400 nm (Fig. 2(c)) demonstrates that all seven elements can be identified in this region. Similarly, all seven elements can be identified in the wavelength region between 350 and 450 nm (Fig. 2 (c and d)).

Armed with this information, it was concluded that univariate analysis is impractical when analyzing complex mixtures such as those reported in this article. Picking the best spectral features from each element is a tedious and time-consuming task, especially if developing

calibration curves for each individual element and then extrapolating that information to the entire mixture of rare earths. In the case of each of the seven elements, six spectra of concentrations each from 1% to 50% were obtained for each rare earth oxides mixed with the balance of graphite powder. Hence partial least square (PLS) was used for the element concentration determination. PLS is a multivariate analysis (MVA) technique that provides a model for the relationship between a set of predictor variables X (n objects, m variables) and a set of response variables Y (n objects, p responses) [49]. In this case, m variables are the LIBS wavelengths and p responses are properties such as the concentrations for the six rare earths and one transition metal. The p response must be independently measured for each sample. If the spectral data contain information about the properties of interest, a reliable calibration model can be constructed. Multivariate analysis of the data is performed using the Unscrambler (version 9.7) software, CAMO, Corvallis, OR [50]. Three important results in using this approach are (1) the r^2 coefficient of determination for the calibration and validation, (2) the loading parameters for each element which should be examined in detail to verify if the features that were selected by the PLS model are real peaks attributed to the specific elements, and (3) the number of principal components (PCs) that the model uses for prediction of those elements. In general, the lower the number of PCs, the better is the development/construction of the model. If the numbers of PCs are greater than 10–15, then there is the danger of integrating on the features in the noise of the spectra and not on the real peaks. A PLS model is used to determine the correlation between measured and predicted rare earth content from the laser-induced breakdown spectra. Fig. 3(a) depicts the predicted versus measured Eu, Gd, La, and Nd scatter plots for the calibration (all samples) and full cross-validation sample sets. Excellent coefficients of determination (r^2) were found for the calibration for each element with 0.97, and 0.95, and 0.97, and 0.98, respectively. Full cross-validation results were slightly lower with 0.95, 0.91, 0.93, and 0.97, respectively.

Fig. 3(b) depicts the predicted versus measured multivariate regression curves for Pr, Sm, and Y scatter plots for the calibration and full cross-validation sample sets. Excellent coefficients of determination (r^2) were found for the calibration sample set for each element with 0.99, 0.97, and 0.99, respectively. Full cross-validation results were slightly weaker (especially for Pr and Sm), with 0.87, 0.89, and 0.97, respectively.

PLS regression models help identify the spectral features present in these rare earth oxide mixtures that contribute to the PLS regression models, in which case acceptable regression coefficients are obtained for both calibration and validation models. No pre-treatment of data was undertaken. Out of the total six spectra that were acquired for each element at each concentration, three were used for calibration and the remaining three were used for validation. PLS1 was performed along with calibration and full cross-validation. PCA was done to find the outliers for each element and those were eliminated from the calibration set. This is shown in Table 2. below.

Fig. 4 shows the loading parameters for all seven rare earth and transition elements that have been identified as the contributing features in distinguishing the rare earth and transition elements between each other.

Table 3 below summarizes the loading parameters for the different elements that the calibration-validation methodology has identified in the wavelength range 390–430 nm.

The first observation is that the PLS1 model selected very few features in the spectra as compared to the original dense, complex features. The prominent loading parameters for these elements are shown in Fig. 4. The spectral features listed in Table 3 are the loading parameters picked out by the program to distinguish the seven elements from each other. The next parameter to be calculated is the total residual variance plot for the seven elements. The total residual

Fig. 5. (a) Distribution of Eu, Gd, La, and Nd variation between the different principal components (PCs). (b) Distribution of Pr, Sm, and Y variation between the different principal components (PCs).

variance plot should be a decreasing function of the number of components. Fig. 5(a) presents the total residual variance plots and is used to assess where the model structure stops and the noise starts for all the correlations. For example, the first graph is the distribution of Eu variation between the different PCs. The second graph in Fig. 5(a) shows the distribution of the variation of Gd data between the different PCs, and the third graph is the distribution of the variation of La data between the different PCs. All the graphs show that most of the variations in the features that contribute to the residual validation model lie within the PC1 components and less variation is found between PC1 and PC2. Even less variation of the features that contribute to the residual validation model is found between PC2 and PC3, and then the variation decreases as expected between PC3 and PC4. This continues until no variation is observed for higher order PCs. The basic PC selection assumes that inclusion of “small” variations do not correspond to significant data structure(s), but rather to noise. That is why it is important to have fewer PCs to develop a robust model.

Similarly in Fig. 5(b), the first graph depicts the PC variation reduced to background for PC = 6, and in the other graphs, the PC number is 4 for samarium and finally for yttrium the PC = 4 at which the variation is completely stabilized.

4. Conclusions

LIBS was used to obtain the complex emission spectra from single-element samples of six rare earth elements Eu, Gd, La, Nd, Pr, and Sm, and one transition metal Y, at concentrations of 1% and 50% by weight for each element in oxide form. Multivariate calibration and cross-validation correlations were developed for each individual element. The calibration r^2 for each element were very good, with values ranging from 0.95 to 0.99. The cross-validation r^2 for each element were slightly lower, with values ranging from 0.87 to 0.97 which is to be expected. A multivariate approach was developed for each element of interest and was able to determine those features that were the primary contributors to the mixed spectra from each specific element. The variation of all the elements between different principal components was found to be between 3 and 6 components, hence confirming that the statistical methodology is very robust and stable. Finally, the loading parameters were confirmed as picked by the PLS multivariate analysis model.

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